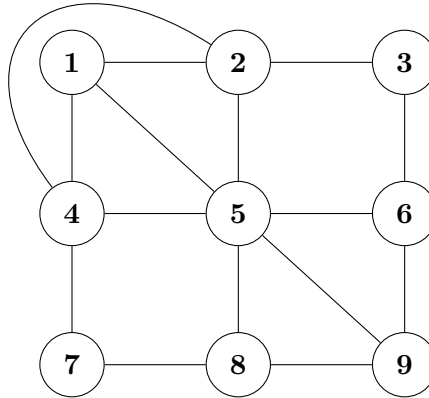


Student ID 1 & 2: (Do this in pairs)

The Graph G . Work with the following graph $G = (V, E)$ on 9 vertices for all four problems. Dashed edges are part of G .



1. Minimum Vertex Cover.

Find a minimum vertex cover of G and argue it is minimum.

2. Maximum Independent Set.

Find a maximum independent set of G .

3. Maximum Clique.

Find a maximum clique of G and show no larger clique exists.

4. Maximum Matching.

Find a maximum matching of G .